Лабораторная работа №8

Настройка SMTP-сервера

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2026-10-14

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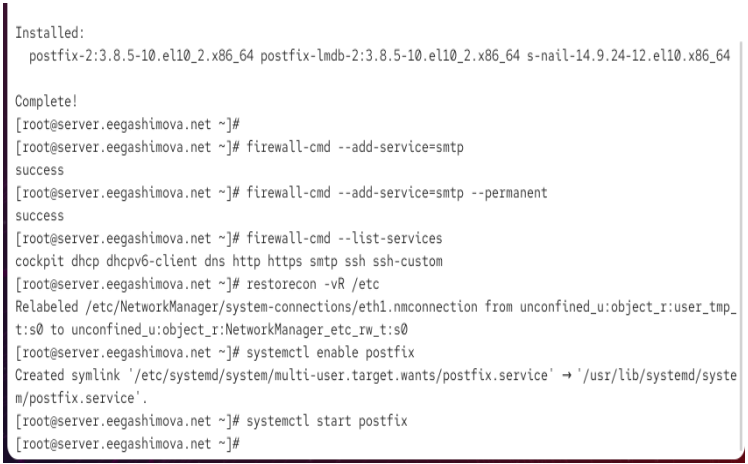
1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Приобретение практических навыков установки, настройки и администрирования почтового сервера **Postfix** в операционной системе Rocky Linux.

Настроить отправку электронной почты между виртуальными машинами **server** и **client**, обеспечить доставку сообщений на доменные адреса и автоматизировать настройку Postfix с помощью Vagrant.

2. 2. Процесс выполнения лабораторной работы



```
Installed:
  postfix-2:3.8.5-10.el10_2.x86_64 postfix-lmdb-2:3.8.5-10.el10_2.x86_64 s-nail-14.9.24-12.el10.x86_64

Complete!
[root@server.eegashimova.net ~]#
[root@server.eegashimova.net ~]# firewall-cmd --add-service=smtp
success
[root@server.eegashimova.net ~]# firewall-cmd --add-service=smtp --permanent
success
[root@server.eegashimova.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https smtp ssh ssh-custom
[root@server.eegashimova.net ~]# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_
t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.eegashimova.net ~]# systemctl enable postfix
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' → '/usr/lib/systemd/syste
m/postfix.service'.
[root@server.eegashimova.net ~]# systemctl start postfix
[root@server.eegashimova.net ~]#
```

Рисунок 1: Установка Postfix, настройка межсетевого экрана и запуск почтовой службы

2.2 Просмотр конфигурации Postfix

```
[root@server.eegashimova.net ~]# postconf
2bounce_notice_recipient = postmaster
access_map_defer_code = 450
access_map_reject_code = 554
address_verify_cache_cleanup_interval = 12h
address_verify_default_transport = $default_transport
address_verify_local_transport = $local_transport
address_verify_map = btree:$data_directory/verify_cache
address_verify_negative_cache = yes
address_verify_negative_expire_time = 3d
address_verify_negative_refresh_time = 3h
address_verify_pending_request_limit = 5000
address_verify_poll_count = ${stress?{1}:{3}}
address_verify_poll_delay = 3s
address_verify_positive_expire_time = 31d
address_verify_positive_refresh_time = 7d
address_verify_relay_transport = $relay_transport
address_verify_relayhost = $relayhost
address_verify_sender = $double_bounce_sender
address_verify_sender_dependent_default_transport_maps = $sender_dependent_default_transport_maps
address_verify_sender_dependent_relayhost_maps = $sender_dependent_relayhost_maps
address_verify_sender_ttl = 0s
address_verify_service_name = verify
address_verify_transport_maps = $transport_maps
address_verify_virtual_transport = $virtual_transport
alias_database = lmdb:/etc/aliases
alias_maps = lmdb:/etc/aliases
```

Рисунок 2: Просмотр текущих параметров конфигурации Postfix

2.3 Изменение параметров myorigin и mydomain

```
[root@server.eegashimova.net ~]#
[root@server.eegashimova.net ~]# postconf myorigin
myorigin = $myhostname
[root@server.eegashimova.net ~]# postconf mydomain
mydomain = eegashimova.net
[root@server.eegashimova.net ~]# postconf -e 'myorigin = $mydomain'
[root@server.eegashimova.net ~]# postconf myorigin
myorigin = $mydomain
[root@server.eegashimova.net ~]# postfix check
[root@server.eegashimova.net ~]# systemctl reload postfix.service
```

Рисунок 3: Проверка и изменение параметров myorigin и mydomain

2.4 Просмотр изменённых параметров Postfix

```
[root@server.eegashimova.net ~]# postconf -n
alias_database = lmdb:/etc/aliases
alias_maps = lmdb:/etc/aliases
command_directory = /usr/sbin
compatibility_level = 3.8
daemon_directory = /usr/libexec/postfix
data_directory = /var/lib/postfix
debug_peer_level = 2
debugger_command = PATH=/bin:/usr/bin:/usr/local/bin:/usr/X11R6/bin ddd $daemon_directory/$process_name
 $process_id & sleep 5
default_database_type = lmdb
html_directory = no
inet_interfaces = localhost
inet_protocols = all
mail_owner = postfix
mailq_path = /usr/bin/mailq.postfix
manpage_directory = /usr/share/man
meta_directory = /etc/postfix
mydestination = $myhostname, localhost.$mydomain, localhost
myorigin = $mydomain
newaliases_path = /usr/bin/newaliases.postfix
queue_directory = /var/spool/postfix
readme_directory = /usr/share/doc/postfix/README_FILES
sample_directory = /usr/share/doc/postfix/samples
sendmail_path = /usr/sbin/sendmail.postfix
setgid_group = postdrop
shlib_directory = /usr/lib64/postfix
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt
smtp_tls_CApath = /etc/pki/tls/certs
smtp_tls_security_level = may
smtpd_tls_cert_file = /etc/pki/tls/certs/postfix.pem
smtpd_tls_key_file = /etc/pki/tls/private/postfix.key
smtpd_tls_security_level = may
unknown_local_recipient_reject_code = 550
[root@server.eegashimova.net ~]#
```

Рисунок 4: Просмотр параметров Postfix, отличающихся от значений по умолчанию

```
[root@server.eegashimova.net ~]#
[root@server.eegashimova.net ~]# postconf -e 'mydomain = eegashimova.net'
[root@server.eegashimova.net ~]# postconf inet_protocols
inet_protocols = all
[root@server.eegashimova.net ~]# postconf -e 'inet_protocols = ipv4'
[root@server.eegashimova.net ~]# postfix check
[root@server.eegashimova.net ~]# systemctl reload postfix.service
[root@server.eegashimova.net ~]#
```

Рисунок 5: Настройка домена eegashimova.net и использование протокола IPv4

2.6 Проверка доставки почты на сервере

```
[eegashimova@server.eegashimova.net ~]$ sudo -i
[sudo] password for eegashimova:
[root@server.eegashimova.net ~]# tail -f /var/log/maillog
Oct 14 08:32:22 server postfix/postfix-script[18521]: refreshing the Postfix mail system
Oct 14 08:32:22 server postfix/master[17219]: reload -- version 3.8.5, configuration /etc/postfix
Oct 14 08:32:22 server postfix/master[17219]: warning: ignoring inet_protocols parameter value change
Oct 14 08:32:22 server postfix/master[17219]: warning: old value: "all", new value: "ipv4"
Oct 14 08:32:22 server postfix/master[17219]: warning: to change inet_protocols, stop and start Postfix
Oct 14 08:33:36 server postfix/pickup[18526]: 2D77222E9B86: uid=1001 from=<eegashimova>
Oct 14 08:33:36 server postfix/cleanup[18715]: 2D77222E9B86: message-id=<20261014083336.2D77222E9B86@se
rver.eegashimova.net>
Oct 14 08:33:36 server postfix/qmgr[18525]: 2D77222E9B86: from=<eegashimova@eegashimova.net>, size=352,
 nrcpt=1 (queue active)
Oct 14 08:33:36 server postfix/local[18719]: 2D77222E9B86: to=<eegashimova@server.eegashimova.net>, rel
ay=local, delay=0.02, delays=0.01/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Oct 14 08:33:36 server postfix/qmgr[18525]: 2D77222E9B86: removed
```

Рисунок 6: Мониторинг журнала Postfix и подтверждение доставки сообщения

2.7 Просмотр локального почтового ящика

```
[eegashimova@server.eegashimova.net ~]$ echo .| mail -s test1 eegashimova@server.eegashimova.net
[eegashimova@server.eegashimova.net ~]$
[eegashimova@server.eegashimova.net ~]$
You have new mail in /var/spool/mail/eegashimova
[eegashimova@server.eegashimova.net ~]$ cat /var/spool/mail/eegashimova
From eegashimova@eegashimova.net  Wed Oct 14 08:33:36 2026
Return-Path: <eegashimova@eegashimova.net>
X-Original-To: eegashimova@server.eegashimova.net
Delivered-To: eegashimova@server.eegashimova.net
Received: by server.eegashimova.net (Postfix, from userid 1001)
        id 2D77222E9B86; Wed, 14 Oct 2026 08:33:36 +0000 (UTC)
Date: Wed, 14 Oct 2026 08:33:36 +0000
To: eegashimova@server.eegashimova.net
Subject: test1
User-Agent: s-nail v14.9.24
Message-Id: <20261014083336.2D77222E9B86@server.eegashimova.net>
From: eegashimova@eegashimova.net

.

[eegashimova@server.eegashimova.net ~]$
```

Рисунок 7: Отправка тестового письма и просмотр содержимого почтового ящика



Рисунок 8: Установка и настройка Postfix на виртуальной машине client

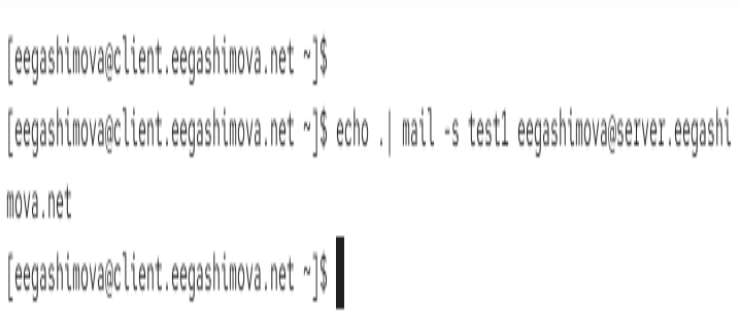


Рисунок 9: Отправка тестового сообщения с клиента на сервер

```
[root@server.eegashimova.net ~]#
[root@server.eegashimova.net ~]# postconf inet_interfaces
inet_interfaces = localhost
[root@server.eegashimova.net ~]# postconf mynetworks
mynetworks = 127.0.0.1/32
[root@server.eegashimova.net ~]# postconf -e 'inet_interfaces = all'
[root@server.eegashimova.net ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
[root@server.eegashimova.net ~]# postfix check
[root@server.eegashimova.net ~]# systemctl reload postfix.service
[root@server.eegashimova.net ~]# systemctl stop postfix.service
[root@server.eegashimova.net ~]# systemctl start postfix.service
[root@server.eegashimova.net ~]#
```

Рисунок 10: Изменение параметров inet_interfaces и mynetworks

2.11 Проверка доставки письма с клиента

```
Oct 14 08:41:30 server postfix/smtpd[20082]: connect from client.eegashimova.net[192.168.1.30]
Oct 14 08:41:30 server postfix/smtpd[20082]: 1E337208294E: client=client.eegashimova.net[192.168.1.30]
Oct 14 08:41:30 server postfix/cleanup[20086]: 1E337208294E: message-id=<20261014084130.13857234B4B8@cl
ient.eegashimova.net>
Oct 14 08:41:30 server postfix/smtpd[20082]: disconnect from client.eegashimova.net[192.168.1.30] ehlo=
2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Oct 14 08:41:30 server postfix/qmgr[20033]: 1E337208294E: from=<eegashimova@client.eegashimova.net>, si
ze=586, nrcpt=1 (queue active)
Oct 14 08:41:30 server postfix/local[20087]: 1E337208294E: to=<eegashimova@server.eegashimova.net>, rel
ay=local, delay=0.01, delays=0/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Oct 14 08:41:30 server postfix/qmgr[20033]: 1E337208294E: removed
```

Рисунок 11: Успешная доставка почтового сообщения с клиента на сервер

2.12 Просмотр полученного сообщения

```
From eegashimova@client.eegashimova.net  Wed Oct 14 08:41:30 2026
Return-Path: <eegashimova@client.eegashimova.net>
X-Original-To: eegashimova@server.eegashimova.net
Delivered-To: eegashimova@server.eegashimova.net
Received: from client.eegashimova.net (client.eegashimova.net [192.168.1.30])
        by server.eegashimova.net (Postfix) with ESMTPS id 1E337208294E
        for <eegashimova@server.eegashimova.net>; Wed, 14 Oct 2026 08:41:30 +0000 (UTC)
Received: by client.eegashimova.net (Postfix, from userid 1001)
        id 13857234B4B8; Wed, 14 Oct 2026 08:41:30 +0000 (UTC)
Date: Wed, 14 Oct 2026 08:41:30 +0000
To: eegashimova@server.eegashimova.net
Subject: test1
User-Agent: s-nail v14.9.24
Message-Id: <20261014084130.13857234B4B8@client.eegashimova.net>
From: eegashimova@client.eegashimova.net

.

[eegashimova@server.eegashimova.net ~]$
```

Рисунок 12: Просмотр письма, полученного сервером от клиентской машины

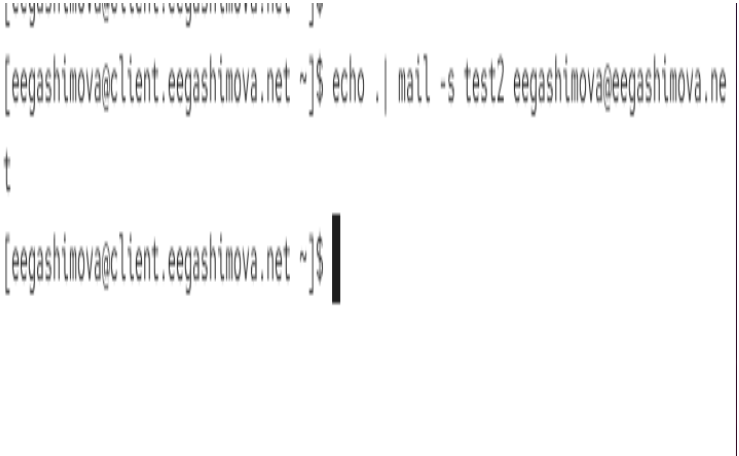


Рисунок 13: Отправка тестового письма на доменный адрес eegashimova.net

2.14 Проверка доставки на доменный адрес

```
Oct 14 08:42:49 server postfix/smtpd[20082]: connect from client.eegashimova.net[192.168.1.30]
Oct 14 08:42:49 server postfix/smtpd[20082]: 766C4208294E: client=client.eegashimova.net[192.168.1.30]
Oct 14 08:42:49 server postfix/cleanup[20086]: 766C4208294E: message-id=<20261014084249.74556234B4B8@cl
ient.eegashimova.net>
Oct 14 08:42:49 server postfix/smtpd[20082]: disconnect from client.eegashimova.net[192.168.1.30] ehlo=
2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Oct 14 08:42:49 server postfix/qmgr[20033]: 766C4208294E: from=<eegashimova@client.eegashimova.net>, si
ze=572, nrcpt=1 (queue active)
Oct 14 08:42:49 server postfix/smtp[20268]: 766C4208294E: to=<eegashimova@eegashimova.net>, relay=none,
 delay=0.01, delays=0/0.01/0/0, dsn=5.4.6, status=bounced (mail for eegashimova.net loops back to mysel
f)
Oct 14 08:42:49 server postfix/cleanup[20086]: 7A38C20E14F0: message-id=<20261014084249.7A38C20E14F0@se
rver.eegashimova.net>
Oct 14 08:42:49 server postfix/qmgr[20033]: 7A38C20E14F0: from=<>, size=2624, nrcpt=1 (queue active)
Oct 14 08:42:49 server postfix/bounce[20269]: 766C4208294E: sender non-delivery notification: 7A38C20E1
4F0
Oct 14 08:42:49 server postfix/qmgr[20033]: 766C4208294E: removed
Oct 14 08:42:49 server postfix/smtp[20268]: connect to client.eegashimova.net[192.168.1.30]:25: No rout
e to host
Oct 14 08:42:49 server postfix/smtp[20268]: 7A38C20E14F0: to=<eegashimova@client.eegashimova.net>, rela
y=none, delay=0, delays=0/0/0/0, dsn=4.4.1, status=deferred (connect to client.eegashimova.net[192.168.
1.30]:25: No route to host)
```

Рисунок 14: Ошибка доставки почтового сообщения на доменный адрес

2.15 Настройка прямой DNS-зоны

```
eegashimova.net    [----]  0 L:[  1+23  24/ 24] *(601 / 601b) <EOF>
$ORIGIN .
$TTL 86400<---->; 1 day
eegashimova.net><------>IN SOA<>eegashimova.net. server.eegashimova.net. (
<------><------><------><------>2026101400 ; serial
<------><------><------><------>86400      ; refresh (1 day)
<------><------><------><------>3600       ; retry (1 hour)
<------><------><------><------>604800     ; expire (1 week)
<------><------><------><------>10800      ; minimum (3 hours)
<------><------><------><------>)
<------><------><------>NS<---->eegashimova.net.
<------><------><------>A<----->192.168.1.1
<------><------><------>MX 10<->mail.eegashimova.net.
$ORIGIN eegashimova.net.
$TTL 1200<----->; 20 minutes
client<><------><------>A<----->192.168.1.30
<------><------><------>DHCID<->( AAEBl6xX1EX/LMyBDkNFiYAawk8eW0jiwP0nX4VKYySh
<------><------><------><------>86Y= ) ; 1 1 32
$TTL 86400<---->; 1 day
dhcp<--><------><------>A<----->192.168.1.1
ns<---><------><------>A<----->192.168.1.1
server<><------><------>A<----->192.168.1.1
www<---><------><------>A<----->192.168.1.1
mail<--><------><------>A<----->192.168.1.1
```

Рисунок 15: Добавление MX-записи и A-записи почтового сервера

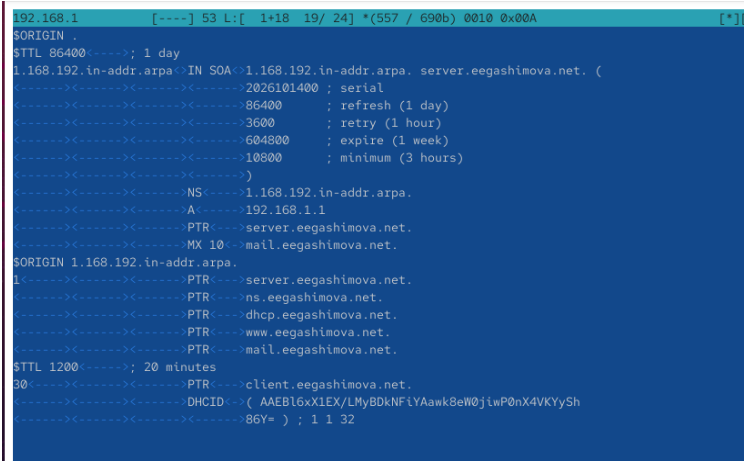


Рисунок 16: Настройка PTR-записей в обратной DNS-зоне

```
[root@server.eegashimova.net ~]#
[root@server.eegashimova.net ~]# postqueue -p
-Queue ID-  --Size-- ----Arrival Time---- -Sender/Recipient-------
7A38C20E14F0    2624 Wed Oct 14 08:42:49  MAILER-DAEMON
        (connect to client.eegashimova.net[192.168.1.30]:25: No route to host)
                                        eegashimova@client.eegashimova.net

-- 2 Kbytes in 1 Request.
[root@server.eegashimova.net ~]# mcedit /var/named/master/fz/eegashimova.net

[root@server.eegashimova.net ~]# mcedit /var/named/master/rz/192.168.1

[root@server.eegashimova.net ~]#
[root@server.eegashimova.net ~]# postconf -e 'mydestination = $myhostname, localhost.$mydomain, localho
st, $mydomain'
[root@server.eegashimova.net ~]# postfix check
[root@server.eegashimova.net ~]# systemctl reload postfix.service
[root@server.eegashimova.net ~]# restorecon -vR /etc
[root@server.eegashimova.net ~]# restorecon -vR /var/named/
[root@server.eegashimova.net ~]# systemctl restart named
[root@server.eegashimova.net ~]# postqueue -f
[root@server.eegashimova.net ~]#
```

Рисунок 17: Проверка почтовой очереди и изменение параметра mydestination

2.18 Проверка доставки после настройки DNS

```
Oct 14 08:49:07 server postfix/smtpd[21349]: connect from client.eegashimova.net[192.168.1.30]
Oct 14 08:49:07 server postfix/smtpd[21349]: 4AB7A208294E: client=client.eegashimova.net[192.168.1.30]
Oct 14 08:49:07 server postfix/cleanup[21351]: 4AB7A208294E: message-id=<20261014084907.45727234B4B8@cl
ient.eegashimova.net>
Oct 14 08:49:07 server postfix/qmgr[21170]: 4AB7A208294E: from=<eegashimova@client.eegashimova.net>, si
ze=572, nrcpt=1 (queue active)
Oct 14 08:49:07 server postfix/smtpd[21349]: disconnect from client.eegashimova.net[192.168.1.30] ehlo=
2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Oct 14 08:49:07 server postfix/local[21352]: 4AB7A208294E: to=<eegashimova@eegashimova.net>, relay=loca
l, delay=0.01, delays=0/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Oct 14 08:49:07 server postfix/qmgr[21170]: 4AB7A208294E: removed
```

Рисунок 18: Успешная доставка сообщения на доменный адрес

```
From eegashimova@client.eegashimova.net  Wed Oct 14 08:49:07 2026
Return-Path: <eegashimova@client.eegashimova.net>
X-Original-To: eegashimova@eegashimova.net
Delivered-To: eegashimova@eegashimova.net
Received: from client.eegashimova.net (client.eegashimova.net [192.168.1.30])
        by server.eegashimova.net (Postfix) with ESMTPS id 4AB7A208294E
        for <eegashimova@eegashimova.net>; Wed, 14 Oct 2026 08:49:07 +0000 (UTC)
Received: by client.eegashimova.net (Postfix, from userid 1001)
        id 45727234B4B8; Wed, 14 Oct 2026 08:49:07 +0000 (UTC)
Date: Wed, 14 Oct 2026 08:49:07 +0000
To: eegashimova@eegashimova.net
Subject: test2
User-Agent: s-nail v14.9.24
Message-Id: <20261014084907.45727234B4B8@client.eegashimova.net>
From: eegashimova@client.eegashimova.net

.
```

Рисунок 19: Просмотр полученного письма с темой test2

```
[root@server.eegashimova.net ~]#
[root@server.eegashimova.net ~]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run-20260920'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/eegashimova.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.ca': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.empty': Protocol error
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.localhost': Protocol err
or
cp: cannot create symbolic link '/vagrant/provision/server/dns/var/named/named.loopback': Protocol erro
r
[root@server.eegashimova.net ~]# cd /vagrant/provision/server/
[root@server.eegashimova.net server]# touch mail.sh
[root@server.eegashimova.net server]# chmod +x mail.sh
[root@server.eegashimova.net server]#
```

Рисунок 20: Копирование файлов DNS и создание сценария mail.sh

2.21 Создание сценария mail.sh на сервере

Vagrantfile | mail.sh

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Copy configuration files"
#cp -R /vagrant/provision/server/mail/etc/* /etc
echo "Configure firewall"
firewall-cmd --add-service=smtp --permanent
firewall-cmd --reload
restorecon -vR /etc
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = eegashimova.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
postfix set-permissions
restorecon -vR /etc
systemctl stop postfix
systemctl start postfix
```

Рисунок 21: Сценарий автоматической установки и настройки Postfix на сервере

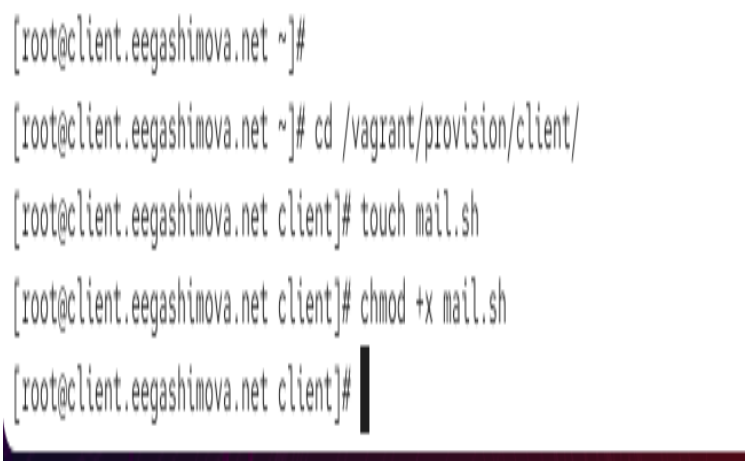


Рисунок 22: Создание исполняемого файла mail.sh на виртуальной машине client

2.23 Создание сценария mail.sh на клиенте

Vagrantfile | mail.sh | mail.sh

```
1  #!/bin/bash
2  echo "Provisioning script $0"
3  echo "Install needed packages"
4  dnf -y install postfix
5  dnf -y install s-nail
6  echo "Configure postfix"
7  postconf -e 'inet_protocols = ipv4'
8  echo "Start postfix service"
9  systemctl enable postfix
10 systemctl start postfix
```

Рисунок 23: Сценарий автоматической установки и настройки Postfix на клиенте

3. 3. Результаты работы

На сервере настроена работа с доменом **eegashimova.net**.

Основные параметры Postfix:

```
mydomain = eegashimova.net
myorigin = $mydomain
inet_protocols = ipv4
inet_interfaces = all
mynetworks = 127.0.0.0/8, 192.168.0.0/16
```

После настройки DNS и Postfix успешно проверена доставка на доменный адрес:

echo . | mail -s test2 eegashimova@eegashimova.net

В журнале **/var/log/maillog** подтверждена успешная доставка сообщений:

status=sent (delivered to mailbox)

4. 4. Вывод

4. Вывод

В ходе лабораторной работы был установлен и настроен почтовый сервер **Postfix** на виртуальных машинах **server** и **client**.

Были настроены параметры SMTP, сетевые интерфейсы, доверенные сети и DNS-записи для домена **eegashimova.net**.

Проверена отправка электронной почты между виртуальными машинами и успешная доставка сообщений на доменный адрес.